Hematology Hemostasis / Coagulation / Platelets Codex Guide

- Block: Hematology
- Method: Codex-authored manual distillation from extracted lecture markdown in this workspace.
- Sources:
  - `2026_Yui_Intro_Platelet_Biology_Primary_Hemostasis.pdf`
  - `20220308_Yui_Quantitative_Platelet_Disorders.pdf`
  - `20220308_Yui_Qualitative_Platelet_Disorders.pdf`
  - `20230306_Yui_Basics_of_the_Coagulation_Cascade.pdf`
  - `2025_Yui_Coagulation_Made_Simple.pdf`
  - `20230306_Yui_Congenital_Acquired_Coagulation_Disorders.pdf`
  - `20230306_Yui_Fibrinolysis_Inhibitors_Coagulation.pdf`
  - `20230306_Yui_Tests_Hemostasis_Thrombosis-1.pdf`

One-Minute Mental Model

Hemostasis has two linked halves. **Primary hemostasis** makes the initial platelet plug using endothelium, vWF, and platelets. **Secondary hemostasis** stabilizes that plug by generating thrombin and fibrin. The body then limits the clot using endogenous anticoagulants and fibrinolysis.

The clinical shortcut:

- **Primary hemostasis problem** -> mucocutaneous bleeding: petechiae, purpura, epistaxis, gingival bleeding, menorrhagia.
- **Secondary hemostasis problem** -> deep bleeding: hemarthroses, muscle bleeds, deep hematomas, delayed bleeding.
- **Platelet count problem** -> thrombocytopenia severity predicts bleeding risk.
- **Platelet function problem** -> mucocutaneous bleeding with normal platelet count.
- **Coagulation factor problem** -> abnormal PT/aPTT pattern; mixing study separates deficiency from inhibitor.

High-Yield Lecture Figures

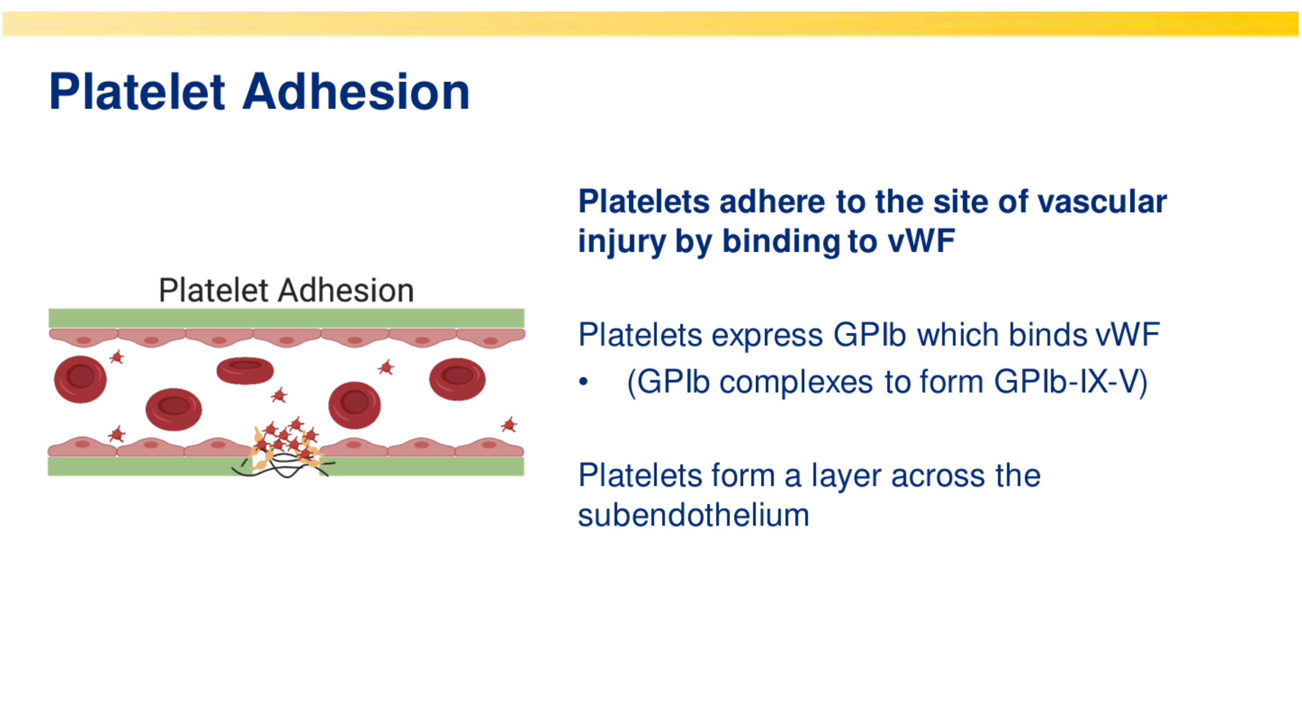

Figure 1: Platelet adhesion through GPIb binding vWF at an injured endothelial surface.

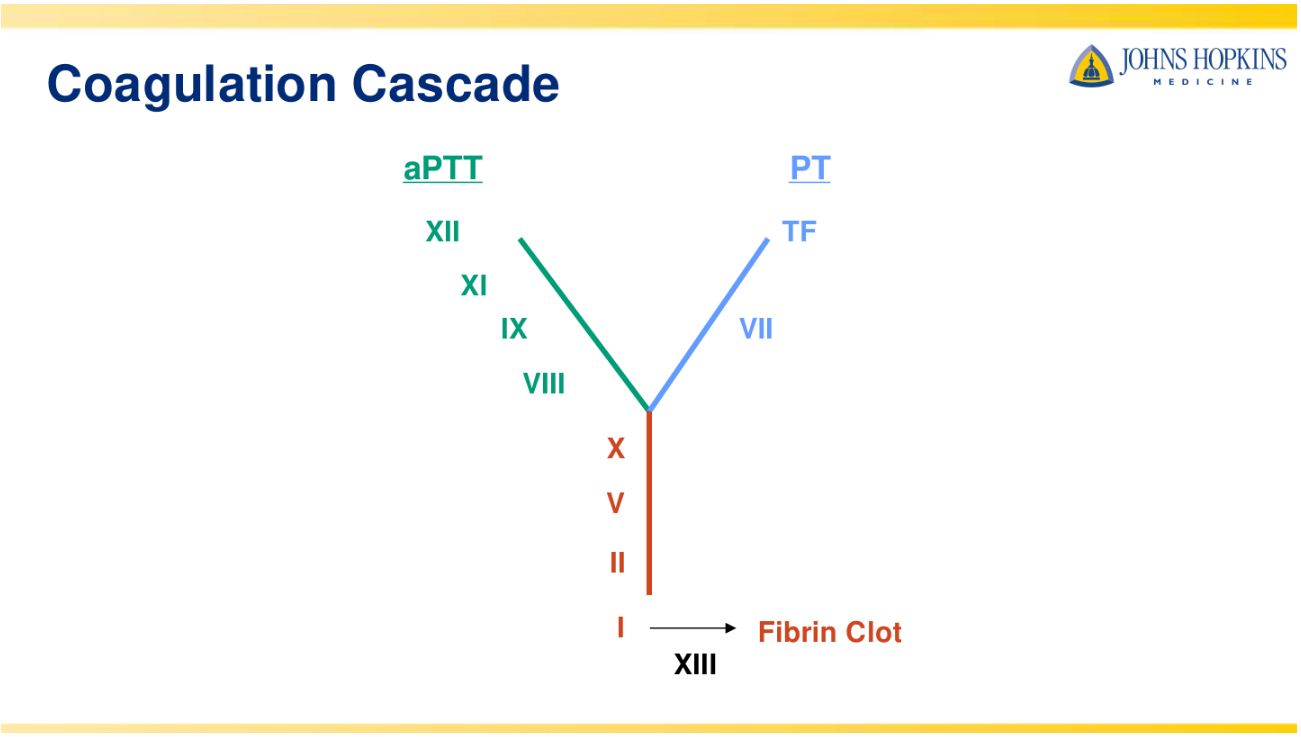

Figure 2: Simplified coagulation cascade showing PT, aPTT, common pathway, and fibrin clot formation.

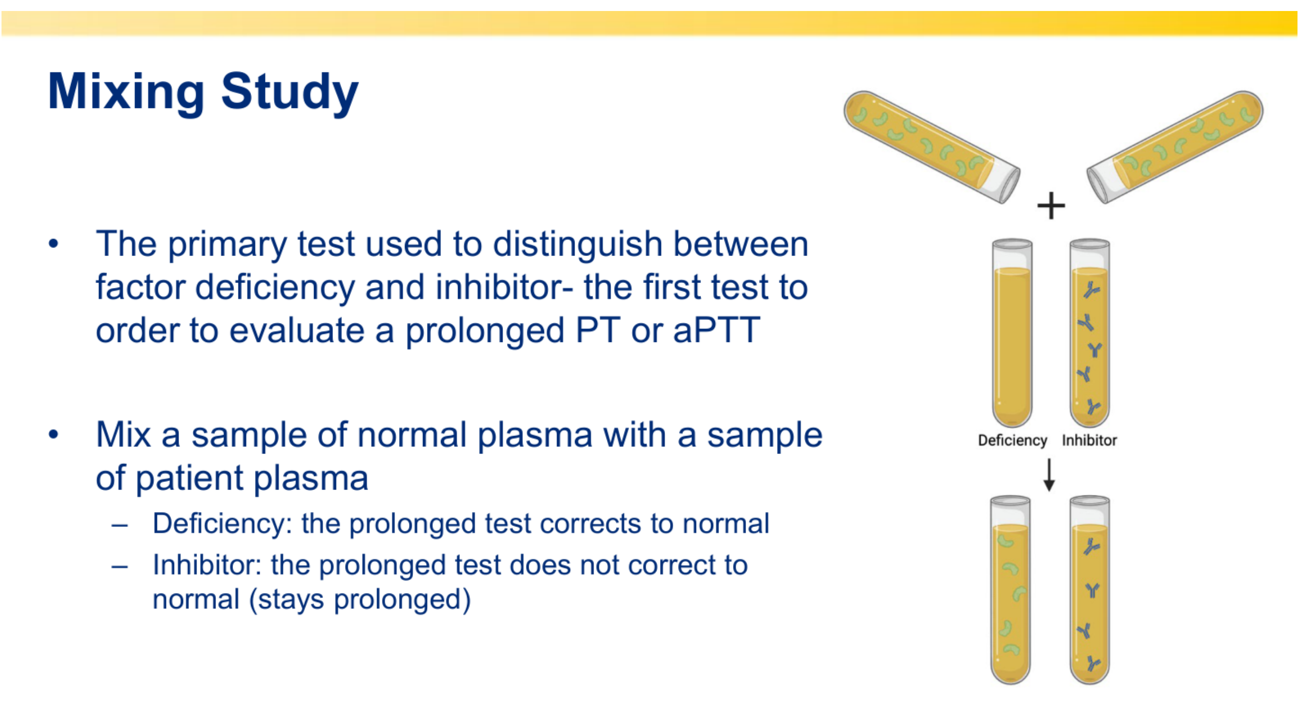

Figure 3: Mixing study logic for separating factor deficiency from inhibitor.

Highest-Yield Concepts

Primary Hemostasis

- Platelets are anucleate cytoplasmic fragments with a 7-10 day lifespan; about one third of platelet mass is stored in the spleen. [Intro Platelet Biology, slide 3]
- Normal endothelium inhibits clotting through nitric oxide and prostacyclin, which promote vasodilation and inhibit platelet adhesion/aggregation. [Intro Platelet Biology, slide 9]
- Endothelial injury releases endothelin and vWF. Endothelin causes vasoconstriction; vWF binds subendothelial collagen and preserves circulating factor VIII. [Intro Platelet Biology, slide 12]
- Platelet adhesion is **GPIb-IX-V on platelets binding vWF** attached to exposed collagen. [Intro Platelet Biology, slide 13]
- Platelet activation causes pseudopodia, phospholipid flipping to the outer surface, degranulation, and release of calcium, ADP, and TXA2. [Intro Platelet Biology, slide 14]
- Alpha granules contain PF4, fibrinogen, factor V, factor XIII, vWF, and platelet-derived growth factor. Dense granules contain calcium, ADP, serotonin, and histamine. [Intro Platelet Biology, slide 6]
- ADP activates platelets via P2Y1/P2Y12 and reveals GPIIb/IIIa; TXA2 promotes platelet activation/aggregation and vasoconstriction. [Intro Platelet Biology, slides 15-16]
- Platelet aggregation is **fibrinogen cross-linking GPIIb/IIIa receptors** on different platelets. [Intro Platelet Biology, slide 17]
- Activated platelets provide phospholipid and calcium support for secondary hemostasis. [Intro Platelet Biology, slide 19; Coagulation Cascade, slides 15-18]

Quantitative Platelet Disorders

- Primary hemostasis defects present with ecchymoses, petechiae, epistaxis, gingival bleeding, and other mucosal bleeding. Secondary hemostasis defects present with joint/muscle bleeding, deep hematomas, and delayed bleeding. [Quantitative Platelet Disorders, slide 3]
- Thrombocytopenia severity: mild 100-150k, moderate 50-100k, severe below 50k; spontaneous hemorrhage becomes more concerning below 20k. [Quantitative Platelet Disorders, slide 7]
- Platelet transfusion is generally reserved for platelets below 10-20k or severe thrombocytopenia with active bleeding; look for reversible marrow suppression. [Quantitative Platelet Disorders, slide 9]
- TPO receptor agonists activate megakaryocyte TPO receptors to increase platelet production. [Quantitative Platelet Disorders, slide 9]
- ITP is IgG autoantibodies against GPIIb/IIIa, causing splenic macrophage clearance. Platelets are low but functionally intact. [Quantitative Platelet Disorders, slide 10]
- ITP often has isolated thrombocytopenia, normal WBC/Hgb, large/giant platelets, and increased megakaryocytes in marrow. [Quantitative Platelet Disorders, slide 12]
- ITP can be primary or secondary to HIV, HCV, SLE, RA, CLL, pregnancy, or other lymphoproliferative disorders. [Quantitative Platelet Disorders, slide 11]
- ITP treatments include corticosteroids, IVIG, rituximab, TPO receptor agonists, and splenectomy. [Quantitative Platelet Disorders, slide 13]
- HIT presents with progressive thrombocytopenia while receiving heparin and/or new thrombosis; it is more common with unfractionated heparin than LMWH. Stop heparin and use a non-heparin anticoagulant. [Quantitative Platelet Disorders, slide 15]
- Thrombocytosis can be primary from MPNs such as essential thrombocythemia, often JAK2/MPL-associated, or reactive from iron deficiency, infection, inflammation, malignancy, autoimmune disease, or post-splenectomy state. [Quantitative Platelet Disorders, slide 16]

Qualitative Platelet Disorders

- Suspect a qualitative platelet disorder when mucocutaneous bleeding occurs without thrombocytopenia. [Qualitative Platelet Disorders, slide 4]
- Acquired platelet dysfunction is most often medication-related, but can also occur with hematologic malignancy, MPNs, myeloma/MGUS, and renal failure/uremia. [Qualitative Platelet Disorders, slide 5]
- Inherited platelet function disorders include Bernard-Soulier syndrome (adhesion defect), storage pool disorders (degranulation/secretion defect), and Glanzmann thrombasthenia (aggregation defect). [Qualitative Platelet Disorders, slide 6]

- Inherited platelet function disorders have mucocutaneous bleeding, family history, normal PT/PTT/fibrinogen, and abnormal platelet function tests. [Qualitative Platelet Disorders, slide 7]
- PFA-100 measures closure time across a collagen-coated membrane with ADP or epinephrine; prolonged closure suggests platelet dysfunction but is not specific. [Qualitative Platelet Disorders, slide 8]
- Platelet aggregometry is the gold standard; agonist panels identify the specific functional defect by light transmission as platelets aggregate. [Qualitative Platelet Disorders, slide 9]
- Ristocetin-induced agglutination tests the vWF-GPIb interaction; abnormal ristocetin response suggests vWF defect or GPIb defect. [Qualitative Platelet Disorders, slide 11]
- Bernard-Soulier is autosomal recessive GPIb-IX-V deficiency: impaired adhesion, moderate thrombocytopenia, large platelets, and absent ristocetin-induced agglutination. [Qualitative Platelet Disorders, slides 12 and 16]
- Glanzmann thrombasthenia is autosomal recessive GPIIb/IIIa deficiency: impaired aggregation, normal platelet count/morphology, absent aggregation to ADP/epinephrine, normal ristocetin response. [Qualitative Platelet Disorders, slides 13 and 16]
- Storage pool disorders impair granule content/release, causing poor mediator release and impaired aggregation. [Qualitative Platelet Disorders, slides 14-15]

Coagulation Cascade

- Coagulation is a series of zymogen-to-active-enzyme reactions; activated factors add "a." The end product is fibrin. [Basics of Coagulation, slides 7-8]
- Vitamin K-dependent factors are **II, VII, IX, X, protein C, and protein S**. Vitamin K enables gamma-carboxylation so factors become functional. [Basics of Coagulation, slide 9]
- The extrinsic/tissue factor pathway is the primary initiation pathway: tissue factor binds VII, forms TF/VIIa, and activates X to Xa. [Basics of Coagulation, slides 10-12]
- The intrinsic pathway starts with contact activation: XII, prekallikrein, and HMW kininogen; XIIa/HMWK activate XI, then XIa activates IX. [Basics of Coagulation, slides 13-15]
- The intrinsic tenase complex is VIIIa + IXa; it activates X to Xa and needs calcium and phospholipid. [Basics of Coagulation, slide 16]
- The common pathway begins at X. Xa + Va is the prothrombinase complex, converting prothrombin II to thrombin IIa. [Basics of Coagulation, slides 17-18]
- Thrombin converts fibrinogen I to fibrin Ia, activates XIII, and amplifies the cascade through positive feedback and platelet activation. [Basics of Coagulation, slides 19-21]
- XIIIa cross-links fibrin monomers into a stable meshwork. [Basics of Coagulation, slide 20]
- Cascade mnemonic: common pathway = X, V, II, I; PT pathway = tissue factor + VII; aPTT pathway = XII, XI, IX, VIII; XIII goes at the end and does not count for PT/aPTT pathway logic. [Coagulation Made Simple, slides 6-12]

Coagulation Disorders and Testing

- Hemophilia A is factor VIII deficiency; hemophilia B is factor IX deficiency. Both are X-linked recessive and cause prolonged aPTT with normal PT; the aPTT corrects on mixing if it is a deficiency. [Congenital/Acquired Coag Disorders, slides 8-9]
- Hemophilia presents with recurrent joint and muscle bleeds, hemarthroses, and hematomas, usually in childhood. [Congenital/Acquired Coag Disorders, slide 8]
- Hemophilia A treatment: factor VIII concentrate, DDAVP, or emicizumab. Hemophilia B treatment: factor IX concentrate. [Congenital/Acquired Coag Disorders, slides 10-13]
- DDAVP releases vWF and factor VIII from endothelial cells and platelets. [Congenital/Acquired Coag Disorders, slide 12]
- Emicizumab is a bispecific antibody that binds IXa to X, functionally replacing VIIIa in the tenase complex. [Congenital/Acquired Coag Disorders, slide 13]
- vWD is the most common inherited bleeding disorder and is usually autosomal dominant. vWF supports platelet adhesion and preserves factor VIII. [Congenital/Acquired Coag Disorders, slides 15-16]
- Mild vWD presents as mucocutaneous bleeding; severe vWD can produce secondary-hemostasis bleeding such as joint/muscle bleeds because factor VIII is low. [Congenital/Acquired Coag Disorders, slide 15]
- vWD tests: vWF antigen, ristocetin cofactor activity, and factor VIII level; severe disease can prolong aPTT. Treatment includes vWF concentrates and DDAVP. [Congenital/Acquired Coag Disorders, slides 17-18]
- Vitamin K deficiency affects II, VII, IX, X, protein C, and protein S; causes prolonged PT and aPTT that correct on mixing; treat with vitamin K. [Congenital/Acquired Coag Disorders, slides 20-21]

- Liver disease decreases production of all coagulation factors except VIII and vWF, causing prolonged PT and aPTT that correct on mixing. [Congenital/Acquired Coag Disorders, slide 22]
- aPTT measures intrinsic + common pathways and is initiated in vitro with factor XII activator, phospholipid, and calcium. [Tests Hemostasis/Thrombosis, slide 7]
- PT measures extrinsic + common pathways and is initiated with tissue factor, phospholipid, and calcium. [Tests Hemostasis/Thrombosis, slide 9]
- Mixing study logic: prolonged PT/aPTT that corrects means factor deficiency; prolonged PT/aPTT that does not correct means inhibitor. [Tests Hemostasis/Thrombosis, slide 11]

Fibrinolysis and Endogenous Anticoagulants

- Fibrinolysis is breakdown of the fibrin clot. Plasmin is the main enzyme and cleaves fibrin into fibrin degradation products; it can also cleave fibrinogen and inhibit platelet aggregation. [Fibrinolysis/Inhibitors, slides 4 and 9]
- tPA from endothelial cells converts plasminogen to plasmin. Alpha-2-antiplasmin inhibits plasmin; PAI-1 inhibits tPA. [Fibrinolysis/Inhibitors, slide 10]
- D-dimer forms when plasmin breaks down cross-linked fibrin while leaving bonds between adjacent D fragments intact. [Fibrinolysis/Inhibitors, slide 17]
- Endothelial cells produce nitric oxide, prostacyclin, and thrombomodulin as antithrombotic mechanisms. [Fibrinolysis/Inhibitors, slide 7]
- Primary endogenous anticoagulants are antithrombin, protein C, and protein S. [Fibrinolysis/Inhibitors, slide 13]
- Antithrombin inhibits thrombin IIa and factors IXa, Xa, and XIa. [Fibrinolysis/Inhibitors, slide 14]
- Thrombin bound to thrombomodulin becomes anticoagulant and activates protein C; protein S is the cofactor for activated protein C. [Fibrinolysis/Inhibitors, slide 15]
- Activated protein C with protein S inactivates VIIIa and Va, blocking tenase and prothrombinase complex formation. [Fibrinolysis/Inhibitors, slide 16]

Must-Know Comparisons

Problem	Typical bleeding	Platelets	PT	aPTT	Key test clue
Thrombocytopenia	Mucocutaneous	Low	Normal	Normal	Count/severity drives risk
Qualitative platelet disorder	Mucocutaneous	Usually normal	Normal	Normal	Abnormal PFA/aggregometry
vWD	Mucocutaneous; severe can be deep	Normal	Normal	Normal/prolonged	Low vWF antigen/activity; +/- low VIII
Hemophilia A	Deep/joint/muscle	Normal	Normal	Prolonged	Low factor VIII; corrects on mix
Hemophilia B	Deep/joint/muscle	Normal	Normal	Prolonged	Low factor IX; corrects on mix
Vitamin K deficiency	Coagulation bleeding	Normal	Prolonged	Prolonged	Corrects on mix; II/VII/IX/X/C/S affected
Liver disease	Coagulation bleeding	Variable	Prolonged	Prolonged	Corrects on mix; all factors except VIII/vWF decreased
Factor inhibitor	Often severe acquired bleeding	Normal	Depends	Depends	Does not correct on mix

Disorder	Defect	Platelet count/morphology	Aggregometry pattern
Bernard-Soulier	GPIb-IX-V adhesion defect	Low count, large platelets	Absent ristocetin response
Glanzmann	GPIIb/IIIa aggregation defect	Normal count, normal morphology	Absent ADP/epinephrine aggregation; normal ristocetin
Storage pool disorder	Granule release/content defect	Often normal	Impaired secondary aggregation
ITP	IgG against GPIIb/IIIa with splenic clearance	Low count, large platelets	Platelets functionally intact
HIT	Heparin-PF4-IgG platelet activation	Low count, normal morphology	High thrombosis risk

Common Confusions To Avoid

- **vWF does two jobs.** It mediates platelet adhesion via GPIb and stabilizes factor VIII.
- **GPIb is adhesion; GPIIb/IIIa is aggregation.** Ristocetin tests vWF-GPIb. Fibrinogen cross-links GPIIb/IIIa.
- **Hemophilia A and severe vWD can both involve low factor VIII.** vWD also has platelet adhesion problems and low vWF activity.
- **PT/aPTT do not test platelet function.** Platelet disorders can have normal PT/aPTT.
- **A mixing study is not optional when PT/aPTT is unexpectedly prolonged.** Correction means deficiency; no correction means inhibitor.
- **Factor XIII is at the end.** It stabilizes fibrin, but it is not part of the PT/aPTT pathway mnemonic.
- **Thrombin is both product and amplifier.** It makes fibrin, activates XIII, activates platelets, and feeds back on other factors.
- **HIT is thrombocytopenia with thrombosis, not bleeding.** Stop heparin and anticoagulate with a non-heparin agent.
- **Vitamin K deficiency affects protein C/S too.** This matters for warfarin physiology and transient hypercoagulability.

Rapid Review

- Primary hemostasis: injury -> endothelin/vWF -> GPIb-vWF adhesion -> platelet activation -> ADP/TXA2/Ca2+ -> GPIIb/IIIa exposure -> fibrinogen-mediated aggregation.
- Secondary hemostasis: TF/VIIa -> X -> Xa/Va -> IIa thrombin -> fibrin -> XIIIa cross-linking.
- Common pathway: X, V, II, I. PT: tissue factor/VII. aPTT: XII, XI, IX, VIII.
- Vitamin K factors: II, VII, IX, X, protein C, protein S.
- Primary bleeding: petechiae, purpura, epistaxis, gingival bleeding, menorrhagia.
- Secondary bleeding: hemarthroses, muscle bleeds, deep hematomas, delayed bleeding.
- ITP: isolated thrombocytopenia, large platelets, increased megakaryocytes, splenic clearance.
- Bernard-Soulier: GPIb defect, large platelets, absent ristocetin.
- Glanzmann: GPIIb/IIIa defect, normal ristocetin, impaired aggregation.
- Hemophilia A/B: prolonged aPTT, normal PT, deep bleeding, corrects on mix.
- vWD: low vWF function/level, mucocutaneous bleeding, possible low VIII/aPTT prolongation.
- Plasmin breaks down fibrin; tPA activates plasmin; D-dimer reflects cross-linked fibrin breakdown.
- Antithrombin inhibits IIa, IXa, Xa, XIa. Protein C/S inhibit Va and VIIIa.

Active Recall Questions

1. Walk through primary hemostasis from endothelial injury to platelet aggregation.
2. What are the two major functions of vWF?
3. Which receptor-ligand pair mediates adhesion? Which mediates aggregation?
4. What is the difference between alpha granule and dense granule contents?
5. How do primary and secondary hemostasis disorders differ clinically?
6. When should you suspect qualitative platelet dysfunction despite a normal CBC?
7. How do PFA-100 and platelet aggregometry differ?
8. How does ristocetin testing distinguish vWF/GPIb pathway problems?
9. Compare Bernard-Soulier and Glanzmann by receptor, platelet morphology, and aggregometry.
10. Why is HIT dangerous even though platelet count is low?
11. Build the coagulation cascade from tissue factor to cross-linked fibrin.
12. Which factors are vitamin K dependent?
13. What does thrombin do besides converting fibrinogen to fibrin?
14. What do PT and aPTT measure?
15. What does correction vs no correction in a mixing study mean?
16. Compare hemophilia A, hemophilia B, and vWD by labs and symptoms.
17. Why can severe vWD cause an aPTT abnormality?
18. How does thrombomodulin change thrombin from procoagulant to anticoagulant?
19. What are the major inhibitors of fibrinolysis?
20. Why does D-dimer imply degradation of cross-linked fibrin rather than just fibrinogen?